

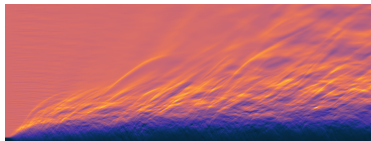
Oceanic Mixed Bottom Layers

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Overview

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Background

Observations in the abyssal ocean show well-mixed layers near the seafloor.

- The Hatteras Abyssal Plain exhibits these bottom mixed layers in the deep ocean. [1]
- Global surveys show especially thick bottom mixed layers in weakly stratified regions. [2]
- The observed layer depth scale with mean current speed, and are affected by background stratification.

In this project, we explore the following question

What physical mechanism produces these thick bottom mixed layer (BML)?

Hatteras Abyssal Plain

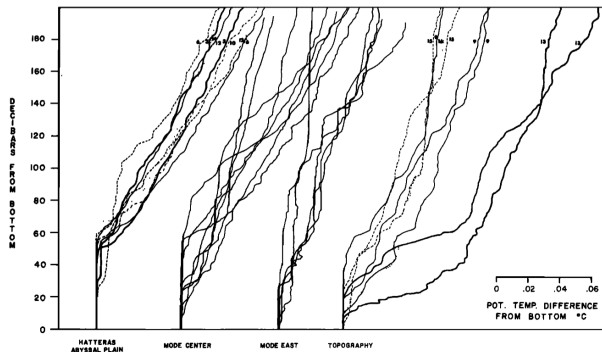


Figure: Potential temperature profiles relative to ocean bottom at various locations around Hatteras Abyssal Plain. [1]

Causes of thick BML

As we have a mixing process to generate turbulence, we investigate potential sources of energy:

- *Benthic storms*: episodic bursts of intense bottom currents, with typical speeds $\sim 20 \text{ cm s}^{-1}$ over weeks
- *Tides*: periodic barotropic tidal current with typical speeds on the order of 1 cm s^{-1} to 2 cm s^{-1} , more important near sloped topographies
- *Wave breaking*: internal wave breaking inducing turbulence

Approximations

- *Boussinesq approximation*: negligible density variations $\rho' \ll \rho_0$, with only momentum contribution to buoyancy
- *Incompressibility*: assume we have incompressible flow

$$\nabla \cdot \mathbf{u} = 0$$

- *Mass conservation/continuity equation*:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad \Longleftrightarrow \quad \frac{D\rho}{Dt} = 0$$

Classical Ekman layer near rigid boundary

Assume far-stream flow $U_\infty \hat{\mathbf{x}}$ in geostrophic balance far from rigid boundary at $z = 0$:

$$fU_\infty = -\frac{1}{\rho} \frac{\partial p}{\partial y}$$

Also, assume vertical flow lengthscale $\ll$ “scale height” of medium $\sim c^2/g$.

Ekman layers form due to balance of *Coriolis* and *friction* forces.

Analytic solution for Ekman layer

With f -plane approximation, where Coriolis parameter $f = 2\Omega \sin \theta$, look for steady-state solution:

$$-fv = \nu \frac{d^2 u}{dz^2}, \quad fu = \nu \frac{d^2 v}{dz^2} + fU_\infty.$$

Boundary conditions: $u = v = 0$ at $z = 0$, and $(u, v) \rightarrow (U_\infty, 0)$ as $z \rightarrow \infty$.

Rewrite $V = u + iv$, analytic solution

$$V = U_\infty \left[1 - \exp \left(-\frac{(1+i)z}{\delta_E} \right) \right]$$

where $\delta_E = \sqrt{2\nu/f}$.

Analytic solution for Ekman layer

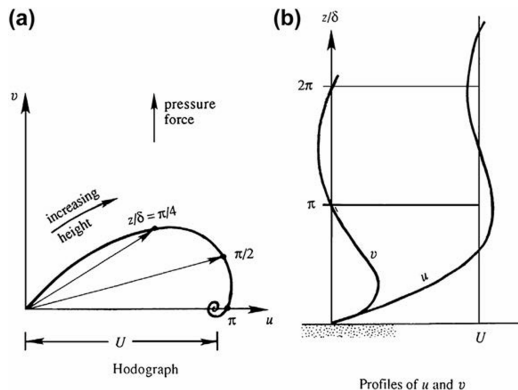


Figure: Plot of analytic solution for Ekman layer at rigid boundary on a hodograph, and velocity components with depth. [3]

Setup for Ekman model

$$\frac{Du}{Dt} - fv = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xz}}{\partial z} \quad (1)$$

$$\frac{Dv}{Dt} + fu = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + fU_{\infty} \quad (2)$$

- Far-stream velocity $U_{\infty} \hat{\mathbf{x}}$

Setup for Stokes model

$$\frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xz}}{\partial z} + U_0 \omega \cos \omega t \quad (3)$$

$$\frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \quad (4)$$

- Far-stream oscillatory velocity $U_0 \hat{x} \sin \omega t$
- No Coriolis effects

Simulating both models

- Simulate bottom mixed layer (BML) evolution for both Ekman and Stokes models
- Use *Oceananigans.jl* in Julia for simulations
- Domains: vertical extent ~ 100 m, time evolution ~ 10 days

Animation

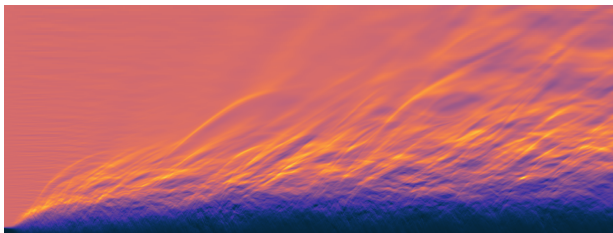


Figure: Ekman flow animation

Key simulation parameters

- *Initial stratification*: uniform density with thin interface above bed
- *Far-stream current*: $U_{\infty} = 0.1 \text{ m s}^{-1}$ (typical abyssal)
- *Stokes drift amplitude*: $U_s(0) = 0.02 U_{\infty}$ at bed (wave contribution)
- *Grid*: high resolution near bottom ($\sim 0.5 \text{ m}$), coarser above
- *Monitoring*: vertical profiles of buoyancy frequency, velocity, and mixed layer depth

Comparing both models

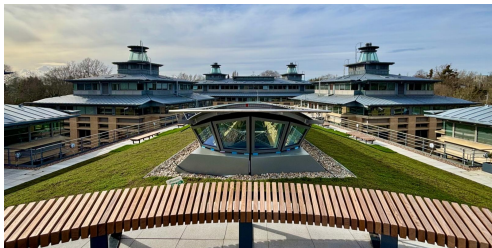
Here's some maths

Summary



Thank you!

Many thanks to our supervisors Prof. John Taylor, Dr Bethan Wynne-Cattanach, and Dr Yuchen Ma, and all the other wonderful people we have been grateful to meet on SRIM! 😊



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